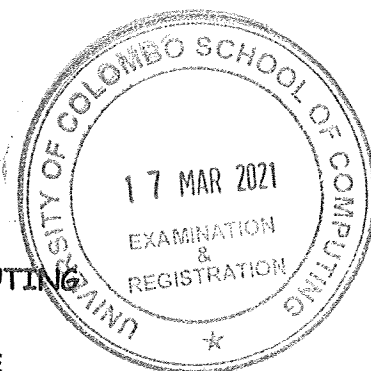




184



UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester II – 2020

SCS1213 – Probability & Statistics

TWO (2) HOURS

To be completed by the candidate

Examination Index No: _____

Important Instructions to candidates:

1. The medium of instruction and questions is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number in each and every page of the question paper.
5. This paper has **4** questions and **11** pages.
6. Answer **ALL** questions. All questions carry equal marks (**25** marks).
7. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are not allowed.
8. Non-programmable calculators are allowed.

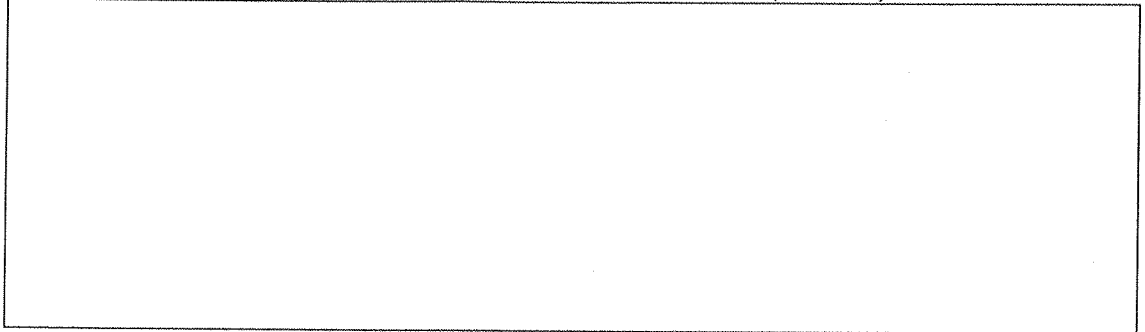
**For Examiner's use
only**

Question No	Marks
1	
2	
3	
4	
Total	

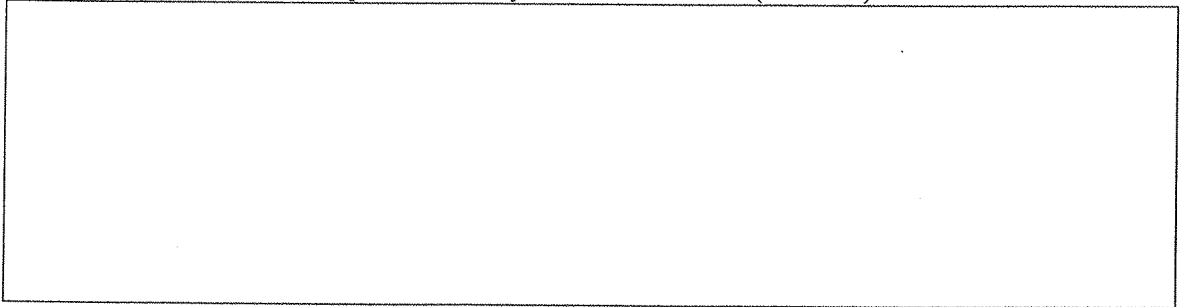
Question 1.

(a) A student sits for two music examinations, theory and practical. He estimates that the probability of passing the theory examination is 0.5, the probability of passing the practical examination is 0.3 and the probability of passing both examinations is 0.2.

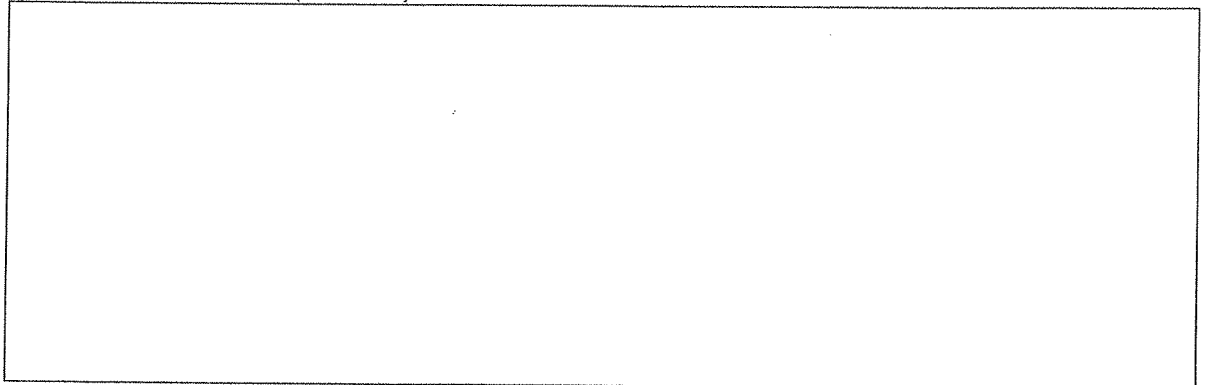
- (i). Find the probability that he does not pass either examination. (4 Marks)



- (ii). Find the probability that he passes exactly one examination. (5 Marks)



- (iii). Given that he does not pass the theory examination, find the probability that he passes the practical examination. (4 Marks)



- (iv). Are the two events, passing theory examination and practical examination independent? Justify your answer. (4 Marks)

- (b) Nizam is playing a game with two friends. The game is designed to be a game of chance so that the probability of anyone winning each game is $\frac{1}{3}$. Nizam and his friends play the game 15 times.

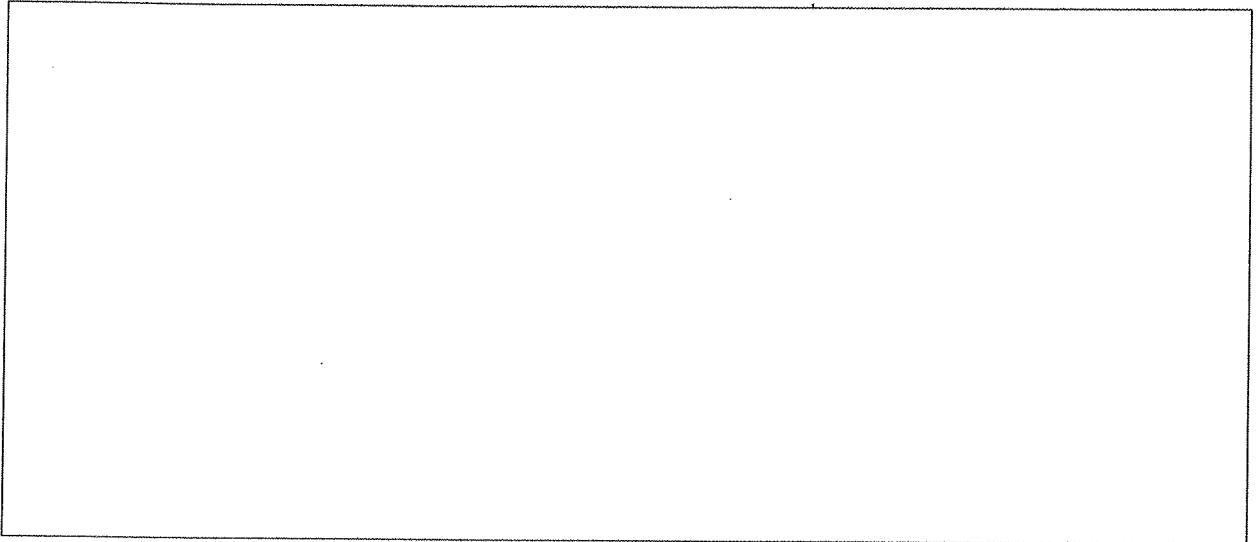
- (i) Find the probability that Nizam wins exactly 2 games. (4 Marks)

- (ii) Find the expected number of games that Nizam will win. (4 Marks)

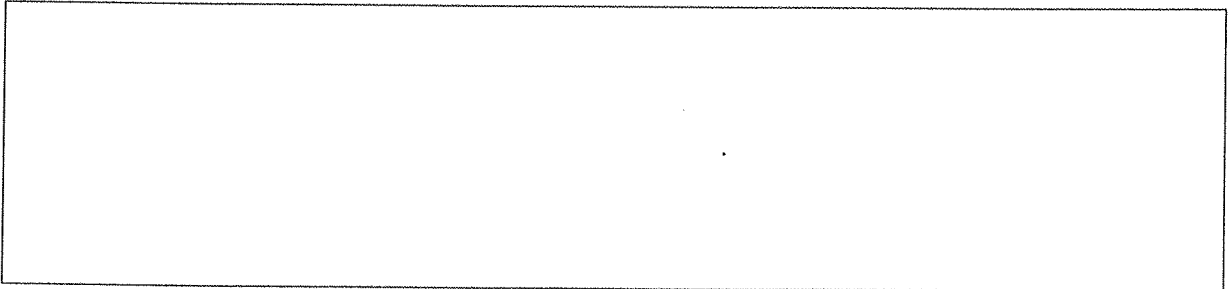
Question 2.

(a) A biased coin with probability of getting head = $\frac{1}{3}$, is tossed 3 times.

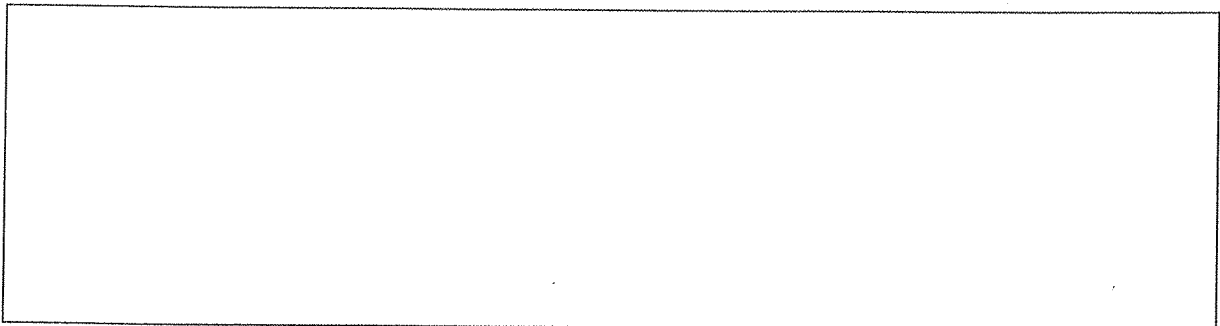
(i) Find the probability that more heads occur than tails. (5 Marks)



(ii) Find the probability that all the outcomes are the same. (3 Marks)

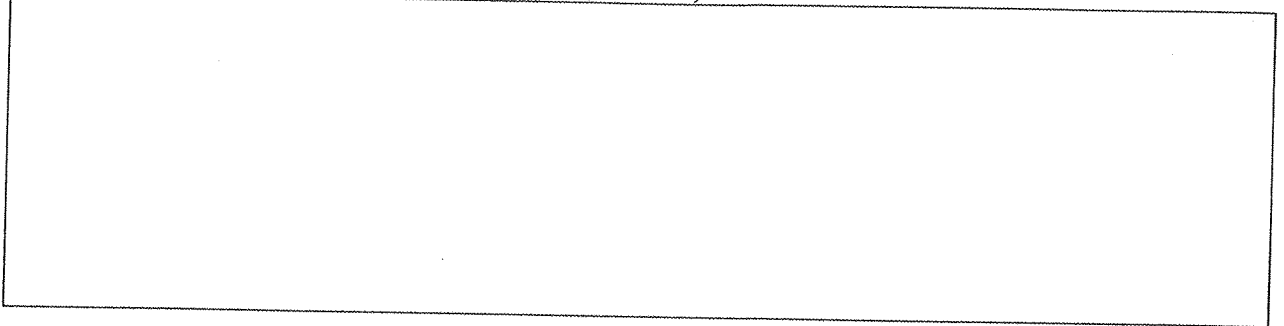


(iii) Find the probability that the first tail occurs on the third throw. (3 Marks)

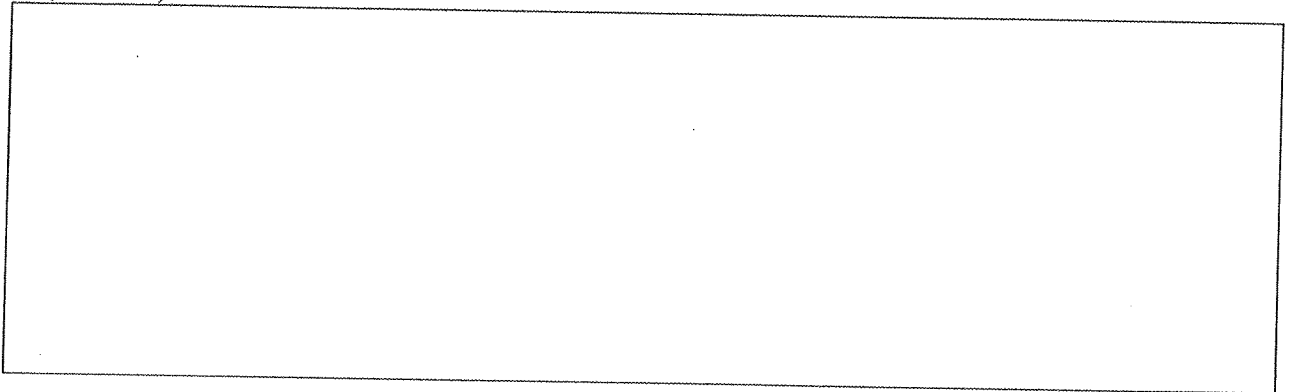


(b) A bag contains a large number of coins. Half of them are \$1 coins, one third are \$2 coins and the remainder are \$5 coins.

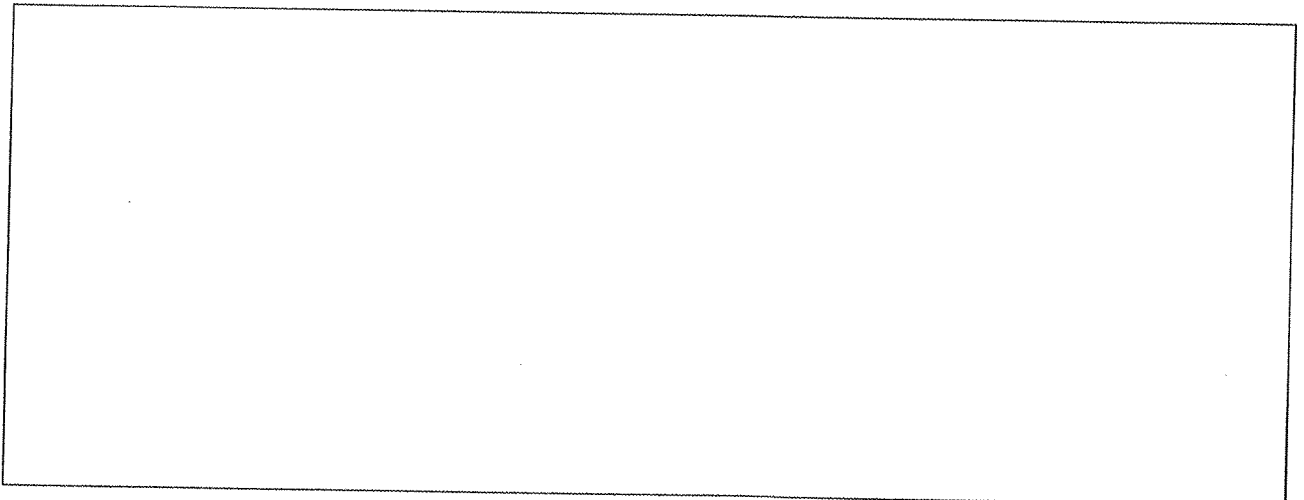
(i) Find the mean of the value of the coins. (3 Marks)



(ii) If a random sample of 2 coins is chosen from the bag, list all possible samples that can be drawn. (3 Marks)



(iii) Let X be the total value of the 2 coins drawn. Find the probability distribution of X . (5 marks)



- (iv) Find the probability that $X < \$12$. (3 Marks)

Question 3.

- (a) The random variable X has probability distribution given in the table below.

x	1	3	5	7	9
$P(X=x)$	0.2	p	0.2	q	0.15

- (i) Given that $E(X) = 4.5$, Find the value of p and q . (6 Marks)

- (ii) Given that $E(X^2) = 27.4$, Find $\text{Var}(X)$. (4 Marks)

- (b) In the manufacture of commercial carpet, small faults occur at random in the carpet at an average rate of 0.95 per 20m^2 . The ground floor of a new office block has 10 rooms. Each room has an area of 80m^2 and has been carpeted using the same commercial carpet described above.

- (i) Determine the probability that in a randomly selected 20m^2 area of this carpet, contains no faults. (3 Marks)

- (ii) For any one of the above described rooms, determine the probability that the carpet in that room contains at least two faults. (4 Marks)

- (iii) Find the probability that in exactly half of these 10 rooms, the carpet will contain exactly 3 faults. (8 Marks)

Question 4.

- (a) The length of rods produced in a workshop is normally distributed with mean μ days and standard deviation σ days. Only 2.5% of all rods are shorter than 235 cm and 15% are longer than 286 cm. Find the value of μ and the value of σ . (7 Marks)

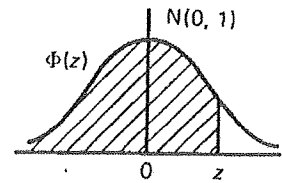
- (b) Anne uses either route A or route B to walk to the university from home every day. Her journey time, X minutes by route A assumed to be normally distributed with a mean of 37 minutes and a standard deviation of 8 minutes. Her journey time, Y minutes by route B assumed to be normally distributed with a mean of 40 minutes and a standard deviation of σ minutes.

- (i) Given that $P(Y > 45) = 0.12$, find the value of σ . (4 Marks)

- (ii) If she leaves home at 8.15am to walk to the university hoping to arrive by 9.00am, which route she should take? Justify your answer. (7 Marks)

- (iii) When Anne travels to the university from home by bicycle, her journey time T minutes follows a normal distribution with mean of 18 minutes and a standard deviation of 12 minutes. For a randomly picked 36 journeys to university from home by bicycle, estimate the probability that, her mean time is greater than 20 minutes. (7 Marks)

THE DISTRIBUTION FUNCTION $\Phi(z)$ OF
THE NORMAL DISTRIBUTION $N(0, 1)$



z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
											ADD								
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359	4	8	12	16	20	24	28	32	36
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753	4	8	12	16	20	24	28	32	36
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141	4	8	12	15	19	23	27	31	35
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517	4	7	11	15	19	22	26	30	34
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879	4	7	11	14	18	22	25	29	32
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224	3	7	10	14	17	20	24	27	31
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549	3	7	10	13	16	19	23	26	29
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852	3	6	9	12	15	18	21	24	27
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133	3	5	8	11	14	16	19	22	25
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389	3	5	8	10	13	15	18	20	23
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621	2	5	7	9	12	14	16	19	21
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830	2	4	6	8	10	12	14	16	18
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015	2	4	6	7	9	11	13	15	17
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177	2	3	5	6	8	10	11	13	14
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319	1	3	4	6	7	8	10	11	13
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441	1	2	4	5	6	7	8	10	11
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545	1	2	3	4	5	6	7	8	9
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633	1	2	3	4	4	5	6	7	8
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706	1	1	2	3	4	4	5	6	6
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767	1	1	2	2	3	4	4	5	5
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817	0	1	1	2	2	3	3	4	4
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857	0	1	1	2	2	2	3	3	4
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890	0	1	1	1	2	2	2	3	3
2.3	.9893	.9896	.9898		.9901	.99036	.99061	.99086			0	1	1	1	1	2	2	2	2
								.99111	.99134	.99158	3	5	8	10	13	15	18	20	23
2.4	.99180	.99202	.99224	.99245	.99266						2	5	7	9	12	14	16	18	21
					.99286	.99305	.99324	.99343	.99361		2	4	6	8	11	13	15	17	19
											2	4	6	7	9	11	13	15	17
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520	2	3	5	6	8	9	11	12	14
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643	1	2	3	5	6	7	8	9	10
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736	1	2	3	4	5	6	7	8	9
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807	1	1	2	3	4	4	5	6	6
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861	0	1	1	2	2	3	3	4	4
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900	0	1	1	2	2	2	3	3	4
3.1	.99932	.99965	.99996								3	6	9	13	16	19	22	25	28
			.999126	.999155	.999184	.999211					3	6	8	11	14	17	20	22	25
3.2	.999313	.999336	.999359	.999381	.999402			.999238	.999264	.999289	2	5	7	10	12	15	17	20	22
					.999423	.999443	.999462	.999481	.999499		2	4	7	9	11	13	15	18	20
3.3	.999517	.999534	.999550	.999566	.999581						2	3	5	6	8	10	11	13	14
					.999596	.999610	.999624	.999638	.999651		1	3	4	5	7	8	9	10	12
3.4	.999663	.999675	.999687	.999698	.999709	.999720	.999730	.999740	.999749	.999758	1	2	3	4	5	6	7	8	9
3.5	.999767	.999776	.999784	.999792	.999800	.999807	.999815	.999822	.999828	.999835	1	1	2	3	4	4	5	6	7
3.6	.999841	.999847	.999853	.999858	.999864	.999869	.999874	.999879	.999883	.999888	0	1	1	2	2	3	3	4	5
3.7	.999892	.999896	.99990	.999904	.999908	.999912	.999915	.999918	.999922	.9999250									
3.8	.999928	.999931	.999933	.999936	.999938	.999941	.999943	.999946	.999948	.9999500									
3.9	.999952	.999954	.999956	.999958	.999959	.999961	.999963	.999964	.999966	.9999670									

For negative values of z use $\Phi(z) = 1 - \Phi(-z)$